Computational strengthenings of Clebsch syndrome rigidity

Tavis Rudd

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Abstract

For a projective arc $A \subset \mathrm{PG}(2, q)$, let $\mathcal{U}(A)$ be the points on no chord of A. The geometric paper proves, without an exhaustive classification of six-arcs over $\mathbb{F}_{11}$, that a six-arc in $\mathrm{PG}(2, 11)$ whose uncovered locus lies on a conic is the Clebsch hexagon. Here exact finite computation sharpens and extends that result. There are fifteen projective classes of six-arcs over $\mathbb{F}_{11}$; the Clebsch class is the unique one whose uncovered locus is contained in a cubic, and it is separated from every other class by a four-point gap in uncovered-set size. A Sylvester-graph obstruction shows that $q = 11$ is the only field order admitting a conic-filling six-arc. Exhaustive orbit searches then classify all conic-filling arcs through eight points: only the projective four-frame over $\mathbb{F}_5$ and the Clebsch six-arc over $\mathbb{F}_{11}$ occur. At the terminal field $q = 13$, Segre tangent triples give a conceptual exclusion of the saturated eight-point case. The associated binary passant/internal incidence code has exact minimum distance 12; its 364 minimum words reconstruct the passant incidence matrix and its full $\mathrm{PGL}(2, 13)$ symmetry. The finite claims are accompanied by exact replay routes and a claim-by-claim trust ledger.

Keywords. finite projective plane; arc; conic; MDS code; deep hole; exhaustive classification.

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1 Setup and relation to the geometric paper

A *k-arc* is a set A of k points of $\mathrm{PG}(2, q)$ with no three collinear. Its chords are the joins of pairs of vertices, and

$$\mathcal{U}(A) = \{P \in \mathrm{PG}(2, q) : P \text{ lies on no chord of } A\}.$$

We call A *conic-filling* when $\mathcal{U}(A)$ is the full rational point set of a nonsingular conic. If n_i counts the off-arc points incident with exactly i chords, put

$$r = \lfloor k/2 \rfloor, \qquad t(A) = \sum_{i=3}^{r} \binom{i-1}{2} n_i, \qquad m = \binom{k}{2}.$$

The geometric paper [9] proves the following three inputs. They are restated to make every dependency in the finite arguments explicit.

Theorem 1.1 (Chord defect [9, Theorem 3.1])**.** *For every k-arc with $k \geq 4$,*

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3\binom{k}{4} - k + 1\right) - t(A), \qquad 0 \leq t(A) \leq 3\binom{k}{4}\frac{r-2}{r}.$$

Theorem 1.2 (Conic-filling window [9, Theorem 3.2]). *If $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and*

$$2k-3 \le q \le \frac{k(k-1)+3}{3} < \binom{k}{2}.$$

For $k=6$, writing $c(A)=n_3$ gives

$$|\mathcal{U}(A)| = q^2-14q+55-c(A), \qquad 0 \le c(A) \le 15, \qquad c(A)=(q-6)(q-9)$$

when $|\mathcal{U}(A)| = q+1$.

Theorem 1.3 (Geometric rigidity [9, Theorem 1.1]). *For a six-arc $A \subset \mathrm{PG}(2,11)$, the locus $\mathcal{U}(A)$ lies on a conic if and only if A is projectively equivalent to the Clebsch hexagon. In that case $\mathcal{U}(A)$ is exactly a nonsingular conic.*

The columns of a parity-check matrix of a projective $[k,k-3,4]_q$ MDS code form a k-arc. When $\mathcal{U}(A)$ is nonempty it is the projective distance-three syndrome locus. Projective equivalence of the unordered column sets is the same as monomial equivalence of the codes. Thus each geometric classification below has the stated coding interpretation.

2 The six-arc census over $\mathbb{F}_{11}$

Computer-assisted conic-inscribed subcensus. For comparison, the exact replay enumerates all 1548 frame-normalized six-arc presentations and selects the 252 whose vertices lie on a nonsingular conic. Their $|\mathcal{U}(A)|$ histogram is

$\lvert\mathcal{U}(A)\rvert$	18	19	20	22
frequency	30	60	90	72

so $|\mathcal{U}(A)| \in \{18,19,20,22\}$. The replay is listed in the "Fifteen classes and numerical gap" row of Table 2.

Table 1 records the complete census needed below. Every representative contains the standard frame

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1);$$

the table gives the remaining two points of each representative. Here $d_{\min}$ is the least degree of a nonzero form vanishing on the uncovered locus.

class	pair	$\lvert G\rvert$	$\lvert\mathcal{U}\rvert$	$d_{\min}$
`C01`	23/32	2	20	5
`C02`	23/34	6	18	4
`C03`	23/36	3	19	5
`C04`	23/37	3	19	5
`C05`	23/39	4	20	5
`C06`	23/42	2	20	5
`C07`	23/48	1	21	5
`C08`	23/49	4	20	5
`C09`	23/56	4	20	5
`C10`	23/62	6	19	5
`C11`	23/67	12	16	4
`C12`	23/74	5	22	6
`C13`	23/89	6	18	4
`C14`	23/(9,10)	12	18	4
`C15`	34/45	60	12	2

Table 1: The fifteen projective classes of six-arcs in $\mathrm{PG}(2,11)$. In the pair column, xy/uv abbreviates $(1,x,y),(1,u,v)$. Class `C15` is the Clebsch class. Both replay programs regenerate the same canonical keys; the global-gap replay supplies $|G|$ and $|\mathcal{U}|$, and the low-degree replay supplies $d_{\min}$.

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 2.1 (Low-degree algebraic rigidity)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q, and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

Computer-assisted proof. For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on $\mathcal{U}(A)$. The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is generated by Q, and multiplication by linear forms gives the full three-dimensional cubic kernel. □

Corollary 2.2 (Monomial characterization)**.** *Up to monomial equivalence, there is a unique linear $[6, 3, 4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. It is the Clebsch code, whose projective stabilizer is A_5.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 2.1 makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. Conversely, the Clebsch code has the stated property, and Dye's Theorem 3 gives the A_5 statement [5, p. 278]. □

Remark 2.3 (Sharpness of the degree threshold). The cutoff cannot be raised from three to four: the four non-Clebsch classes `C02`, `C11`, `C13`, and `C14` have $d_{\min} = 4$, so the failure is not isolated. For `C02`, the low-degree replay in Table 2 prints a quartic whose rational zero set is exactly the 18-point uncovered locus and verifies the equality point by point. No smoothness, irreducibility, or genus assertion is used here.

Theorem 2.4 (Numerical gap)**.** *The Clebsch class has $|\mathcal{U}(A)| = 12$, while every non-Clebsch six-arc has $|\mathcal{U}(A)| \geq 16$. Equivalently, $|\mathcal{U}(A)| \leq 15$ characterizes the Clebsch class, for which $|\mathcal{U}(A)| = 12$. If $|\mathcal{U}(A)| \geq 16$, then $\mathcal{U}(A)$ lies on no conic.*

Computer-assisted proof. Any ordered four points of a six-arc form a projective frame, uniquely normalized to $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$. Enumerating the remaining unordered pair subject to the arc condition gives 1548 normalized sets. Each embedded arc has $\binom{6}{4} 4! = 360$ ordered-frame choices. For each choice, apply the unique normalizing projectivity, sort the resulting six normalized points, and encode that list lexicographically. Taking the least of the 360 lists removes the frame choice; two arcs have the same least list exactly when a projectivity carries one to the other. This canonical key gives fifteen classes, and the number of distinct normalized presentations of a class with stabilizer G is $360/|G|$. Their extension counts are

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\},$$

with multiplicities

$$(6, 30, 150, 300, 630, 360, 72)$$

across the 1548 normalized representatives. The six presentations attaining 12 form the single Clebsch class; every other class has at least 16 extension points. Theorem 1.3 supplies the final conic-containment assertion. ☐

3 Cross-field uniqueness for six-arcs

Lemma 3.1 ($q = 9$ polarity graph)**.** *For a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2,9)$, let Σ be the graph on the 36 internal points in which*

$$P \sim Q \quad \Longleftrightarrow \quad Q \in P^{\perp}$$

for the conic polarity. Then Σ is the Sylvester graph, with intersection array

$$\{5, 4, 2; 1, 1, 4\}.$$

For distinct internal points P, Q, the line PQ is passant to $\mathcal{C}$ if and only if P and Q are at distance two in Σ, and the graph joining pairs at distance two has clique number

$$\omega(\Sigma_2) = 5.$$

Proof. Counting internal points on polar lines and their intersections gives $b_0 = 5$, $b_1 = 4$, $b_2 = 2$, $c_1 = c_2 = 1$, and $c_3 = 4$, hence the displayed intersection array. This is the Sylvester graph [3, §13.1.2]; uniqueness for this intersection array is proved in [7, Thm. 6].

For distinct internal points P, Q, put $R = P^{\perp} \cap Q^{\perp} = (PQ)^{\perp}$. If PQ is passant, then its pole R is internal and adjacent to both P and Q; since the intersection array has $a_1 = 0$, these vertices are at distance two. Conversely, a common neighbor of P and Q must equal R, so distance two makes the pole of PQ internal and hence makes PQ passant. The final value is $\mathrm{eq}_2(\Sigma) = 5$ in [1, Table 1]. ☐

Theorem 3.2 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2,q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. At $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Theorem 1.2 gives $q \geq 9$ and $c(A) = (q-6)(q-9)$ with $0 \leq c(A) \leq 15$. Thus the only prime-power candidates are $q = 9, 11$: $q = 10$ is not a prime power, and every $q \geq 12$ gives $c(A) \geq 18$.

It remains to exclude $q = 9$. Since every point of $\mathcal{C}$ is uncovered, no chord of A meets $\mathcal{C}$; every chord is therefore passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external, so the five chords through each vertex force all six vertices of A to be internal. Lemma 3.1 would then make A a six-clique in the distance-two graph of the Sylvester graph, whose clique number is five. This is impossible.

Therefore $q = 11$. The final assertion follows from Theorem 1.3, which puts every conic-filling six-arc over $\mathbb{F}_{11}$ in the single Clebsch orbit. ☐

4 The binary passant code at $q = 13$

Fix the conic $\mathcal{C} : XZ - Y^2 = 0$ in $\mathrm{PG}(2,13)$. Let M be the 78-by-78 incidence matrix over $\mathbb{F}_2$, with rows indexed by the passant lines to $\mathcal{C}$ and columns indexed by its internal points. Madison and Wu determine the nullity of this matrix for every odd q [8, Corollary 6.3]; their formula $(q-1)^2/4$ gives 36 at $q = 13$. Hollmann and Xiang construct the elliptic association scheme afforded by the $\mathrm{PGL}(2,q)$-action on the passant carrier [6, §3]. Thus the dimension and the association scheme used below are known. The assertions here are the exact minimum distance and the intrinsic reconstruction of the incidence matrix and scheme from the minimum words.

Theorem 4.1 (Tangent-code theorem). *The binary code $K = \ker M$ has parameters*

$$[78, 36, 12]_2.$$

There are exactly 364 *minimum words. Under* $\mathrm{PGL}(2,13)$ *they form four orbits of size* 91*, one with stabilizer* S_4 *and three with stabilizer* D_{24}*. Each orbit spans* K*.*

The minimum supports intrinsically reconstruct the full six-class elliptic association scheme on the 78 *internal points. Pair concurrence recovers passant versus secant join type, triple concurrence separates the six orbitals, and the* 78 *all-zero-triple seven-cliques are exactly the passant incidence rows. Consequently the minimum layer reconstructs* M *up to row order, and its automorphism group is* $\mathrm{PGL}(2,13)$*.*

Proof. Every row and column of M has weight seven, and conic polarity identifies the two index sets. Madison and Wu's dimension formula gives $\dim K = 36$, equivalently $\operatorname{rank} M = 42$; exact elimination independently checks this specialization. Summing the row equations gives the all-one functional, so every word of K has even weight. Moreover, if a nonzero word contains an internal point P, each of the seven passants through P must contain another support point. These seven points are distinct, proving the elementary lower bound $\mathrm{wt} \geq 8$.

Suppose first that a word has weight eight. The preceding parity argument shows that its support is an eight-arc, every join of two support points is passant, and the passant pencil at every vertex is saturated. Thus the seven remaining lines through each vertex are precisely the seven combinatorial tangents to the arc and the seven conic-secants through that internal point.

For an internal point P, let

$$T_P(X) = \prod_{\substack{\ell \ni P \\ \ell \text{ secant to } \mathcal{C}}} \ell(X).$$

For a pairwise-passant triple put

$$h(P,Q,R) = \frac{T_P(Q)T_Q(R)T_R(P)}{T_P(R)T_R(Q)T_Q(P)}.$$

Segre's lemma of tangents [2, Lemma 27] gives $h(P,Q,R) = 1$ for every triple in the hypothetical support. Fixing $P = (1:0:2)$ therefore makes its other seven points a clique in the graph on the 42 passant-join neighbors of P, where Q and R are adjacent when QR is passant and $h(P,Q,R) = 1$.

An order-14 element of the stabilizer of P splits these vertices into three cyclic orbits A_i, B_i, C_i, with indices in $\mathbf{Z}/14$. Direct substitution in the displayed product gives the complete adjacency rule

orbit pair	$j - i \pmod{14}$
AA	$4, 6, 8, 10$
AB	$6, 7, 11, 12$
AC	$1, 3$
BB	$6, 8$
BC	$3, 5, 6, 8, 9, 11$
CC	$2, 4, 10, 12.$

Up to simultaneous translation, every four-clique is one of five rows:

four-clique	unique extension
A_0, B_6, B_{12}, C_1	C_3
A_0, B_6, B_{12}, C_3	C_1
A_0, B_6, C_1, C_3	B_{12}
A_0, B_{12}, C_1, C_3	B_6
B_0, B_6, C_9, C_{11}	A_8.

They close to fourteen translates of one five-clique, none of which extends. The local clique number is therefore five, not seven. This excludes weight eight.

For weight ten, fix one support point. Parity leaves only two passant-pencil profiles: three points on one passant and one on each of the other six, or one on each passant together with two secant-join neighbors. Exact meet-in-the-middle syndrome checks exhaust respectively

$$7\binom{6}{3}6^6 = 6{,}531{,}840, \qquad 6^7\binom{35}{2} = 166{,}561{,}920$$

supports, and neither profile has zero syndrome. Finally, the following twelve internal points do have zero syndrome:

$$\begin{gathered}(1:0:2), (1:3:2), (1:4:5), (1:1:8),\\ (1:4:8), (1:1:7), (1:7:12), (1:3:3),\\ (1:9:11), (1:10:11), (1:0:5), (1:8:7).\end{gathered}$$

Their stabilizer is dihedral of order 24. This proves $d(K) = 12$.

The remaining assertions are an exhaustive exact classification of the minimum layer. The replay generates the four projective orbits from explicit representatives, verifies all 364 supports against M, and computes their binary spans. It then forms all pair and triple concurrence profiles. Pair concurrences are $7, 9, 12$ on passant joins and $6, 8$ on secant joins; the six triple histograms are distinct. Exactly 78 of the resulting passant seven-cliques have zero triple concurrence, and direct comparison identifies them with the rows of M. A stabilizer-chain enumeration of permutations preserving these relations has order $2184 = |\operatorname{PGL}(2, 13)|$, completing the reconstruction and automorphism claims. $\square$

5 The small-arc boundary

The chord moments reduce the neighboring cases through eight points to a finite list of fields. Exact terminal enumeration excludes the remaining seven- and eight-point cases.

Theorem 5.1 (Conic-filling loci through eight points)**.** *Let A be a k-arc in* $\mathrm{PG}(2, q)$*, where q is a prime power and $4 \le k \le 8$, and suppose that $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$-rational points of a nonsingular conic. Then*

$$(k, q) = (4, 5) \qquad \text{or} \qquad (k, q) = (6, 11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur.

Moreover, for each $q \in \{13, 17, 19\}$, an arc all of whose chords are passant to a fixed nonsingular conic has at most six points, and this bound is attained.

Proof with exhaustive finite verification. For $k = 4$, Theorem 1.2 gives $5 \le q < 6$, hence $q = 5$. Every four-arc is a projective frame, and for the standard frame a direct substitution gives

$$\mathcal{U}(A) = V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

where $V(f)$ denotes the projective zero set of f; this is a nonsingular six-point conic. For $k = 5$, Theorem 1.1 gives $|\mathcal{U}(A)| = q^2-9q+21$; equality with $q+1$ would require $q^2-10q+20 = 0$, which has no integral root. The case $k = 6$ is Theorem 3.2 together with Theorem 1.3.

For $k = 7$, Theorem 1.1 gives

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q + 1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The strengthened conic-filling window gives $11 \le q \le 15$. Thus only the prime powers $q = 11, 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized enumeration now supplies the finite exclusion. It finds 10232 distinct normalized seven-arcs at $q = 11$, of which 140 have $|\mathcal{U}(A)| = q+1$, and 53960 at $q = 13$, of which 1680 have that size. Every seven-arc is reached because it contains a four-frame; extending each normalized six-arc by its uncovered points produces every resulting seven-arc exactly three times, according to which non-frame point is deleted. For all 1820 candidates of the required size, the quadratic evaluation matrix has full rank six, so none lies on a conic.

Finally let $k = 8$. The strengthened conic-filling window gives $13 \le q \le 59/3$, so the odd prime powers are exactly $13, 17, 19$.

It remains to exclude these three fields. By projective equivalence fix the conic $XZ = Y^2$, whose stabilizer is the symmetric-square image of $\mathrm{PGL}(2, q)$. Its complement has q^2 points. In the terminology of Blokhuis, Seress, and Wilbrink, a set whose pairwise joins are passant is an exterior set of the conic [4, pp. 143,146]; its vertices may have either internal or exterior point type. Every conic-filling arc is such an exterior set.

There is already a structural reduction at $q = 13$. An exterior point lies on only $(q - 1)/2 = 6$ passants, fewer than the seven distinct chords through a vertex of an eight-arc. Hence every vertex would be internal. Each internal point lies on exactly $(q+1)/2 = 7$ passants, so its seven chords would exhaust its complete passant pencil. Its characteristic vector would therefore be a weight-eight word of the code in Theorem 4.1, which is impossible. This gives the conceptual $q = 13$ exclusion needed for conic filling.

The stronger assertion that every pairwise-passant arc has at most six points still needs the orbit search. Join two off-conic points when their line is passant. The desired eight-arc would be an eight-clique in this graph, with collinear triples forbidden.

An exact orbit search gives the following data:

q	off-conic points	passant edges	$\mathrm{PGL}(2, q)$-orbits on edges
13	169	7098	10
17	289	20808	13
19	361	32490	15.

For one representative of each edge orbit, the search extends partial arcs and identifies states under the setwise stabilizer of the root edge. Canonical keys are the lexicographically least sorted point lists in each stabilizer orbit. Every candidate is reached: it contains a passant edge, that edge is carried to one of the listed representatives, and the remaining points are then among the enumerated extensions. In all three fields the largest partial arc has size six. Thus there is no seven-point arc with all chords passant, and in particular no conic-filling eight-arc. A separate replay verifies the edge-orbit partition and repeats the extension search by ordered backtracking without stabilizer identification. Its certificates give an explicit six-point witness in every field, proving sharpness. □

In coding terms, let C be a projective $[k, k-3, 4]_q$ MDS code with $4 \leq k \leq 8$. If its projective distance-three syndrome locus is a nonsingular conic, then, up to monomial equivalence, C is the $[4, 1, 4]_5$ frame code or the $[6, 3, 4]_{11}$ Clebsch code. Both codes have the stated syndrome locus.

Thus, for each fixed arc size, only finitely many field orders can admit conic filling; the classification is complete through eight points.

6 Trust and exact replay

The claims in this paper use three distinct proof modes. The chord-defect, window, and rigidity inputs are proved in the geometric paper. The $q = 9$ graph identification and clique bound use the cited literature, as does the tangent identity in the conceptual $q = 13$ reduction. The census, rank calculations, and terminal orbit searches are exhaustive finite computations. Table 2 lists the load-bearing executable route for each such claim; these checks use exact finite-field arithmetic and integer combinatorics, not randomized tests.

The machine-readable version of this ledger is the file

`verification/computational_companion_trust.json`.

Its paths are relative to the present directory. Recorded outputs and source hashes are in `verification/checker_outputs.json`; the larger $k = 8$ certificates are checked by `verification/conic_filling_verify.py`. The clean companion entry point is

`verification/verify_computational_companion.py`.

Passing `--run` also executes every replay in Table 2.

Result	Exhaustive or independent executable check
Fifteen classes and numerical gap	`check_global_conic_gap.py`
Low-degree rigidity and quartic witness	`check_low_degree_loci.py`
$q = 9$ exclusion and uniqueness of $q = 11$	`check_small_q_uniqueness.py`
Seven-arc exclusions at $q = 11, 13$	`check_small_k_conic_filling.py`
$q = 13$ tangent code and minimum layer	`check_q13_tangent_code.py`
Passant-arc bound for $q = 13, 17, 19$	`verification/conic_filling_verify.py`

Table 2: Load-bearing exact replays for the computational claims.

No exhaustive classification in this paper is certified by Lean. A separate development checks the chord moments, six-arc defect specialization, the $q = 9$ polarity reduction, and the geometric bridge used in the small-k argument. Accordingly, the replays in Table 2 are part of the proof surface rather than optional corroboration.

7 Conclusion

The finite census exposes two strengthenings that the geometric proof does not need: degree-three containment already characterizes the Clebsch class, and uncovered-set size has a four-point gap above it. Across fields, the Sylvester obstruction makes $q = 11$ unique for conic-filling six-arcs. Together with the terminal searches, this leaves exactly two conic-filling projective codes through length eight. At the first terminal field, the tangent-code argument also identifies the exact binary distance and shows that the complete minimum layer reconstructs its passant geometry.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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